

Stochastic Gradient Descent

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Gradient descent ([Note 57](#)).
- **Leads to:** Perceptron loss ([Note 1006](#)); Batch size in Keras ([Note 1020](#)).
- **Compare with:** Batch gradient descent ([Note 58](#)); Mini-batch gradient descent ([Note 60](#)).

1. Overview

Key point: Stochastic gradient descent updates the coefficients after every single random observation. Each step is cheap and noisy, so it gets close to the answer in far fewer epochs than batch gradient descent, but it jitters around the minimum.

A **feature** is an input variable (one column of the data table), an **observation** is one record (one row), and the **target** is the output we predict.

Batch gradient descent, from the previous Note, uses every observation of the training data for each update. **Stochastic gradient descent (SGD)** uses just one observation, picked at random, for each update. SGD and its variants are probably the most used optimisation algorithms in machine learning, and nearly all of deep learning is trained with it (Goodfellow §8.3.1, §5.9).

This Note explains the problem with batch gradient descent, how SGD works, its behaviour, and how to use it in scikit-learn.

2. The problem with batch gradient descent

Key point: Every batch update touches every observation and every feature. With big data, every step is a huge amount of work, and the whole dataset must sit in memory.

To update once, batch gradient descent computes the derivative for every coefficient, and each derivative sums over all n observations. So one epoch costs about $n \times m$ multiplications, and the whole run costs about $n \times m \times \text{epochs}$.

With numbers: a dataset of 100,000 observations and 100 features, trained for 1,000 epochs, needs about

$$100,000 \times 100 \times 1,000 = 10^{10}$$

multiplications, ten billion, just for the derivatives.

So each step is slow and needs the whole dataset in memory: the two disadvantages listed in the [batch gradient descent Note](#) (section six).

3. How stochastic gradient descent works

Key point: In each epoch, repeat n times: pick one random observation, compute the derivatives from that observation alone, and update.

The word **stochastic** means random. The algorithm:

1. Start with any coefficients, for example $\beta_0 = 0$ and every $\beta_j = 1$.
2. For each epoch, repeat n times:
 - pick one observation i at random;
 - compute its prediction $\hat{y}_i$ and error $y_i - \hat{y}_i$;
 - update every coefficient using the derivatives from that one observation.
3. Stop after the chosen number of epochs.

The derivatives are the batch ones with the sum over observations removed: only observation i remains.

$$\frac{\partial L}{\partial \beta_0} = -2(y_i - \hat{y}_i) \quad \frac{\partial L}{\partial \beta_j} = -2(y_i - \hat{y}_i) x_{ij}$$

With numbers: if observation i has error $y_i - \hat{y}_i = 5$ and $x_{ij} = 0.04$, then $\partial L / \partial \beta_j = -2 \times 5 \times 0.04 = -0.4$, and with learning rate 0.01 the coefficient rises by 0.004.

So an epoch of SGD makes n small updates instead of one big one. On the diabetes data an epoch is 353 updates per epoch.

Python: Stochastic gradient descent from scratch.

```
class MySGDRegressor:
    def __init__(self, learning_rate=0.01, epochs=100, seed=0):
        self.lr, self.epochs = learning_rate, epochs
        self.rng = np.random.default_rng(seed)

    def fit(self, X, y):
        n = X.shape[0]
        self.intercept_, self.coef_ = 0.0, np.ones(X.shape[1])
        for _ in range(self.epochs):
            for _ in range(n):
                i = self.rng.integers(n)          # one random row
                err = y[i] - (X[i] @ self.coef_ + self.intercept_)
                self.intercept_ -= self.lr * (-2 * err)
                self.coef_ -= self.lr * (-2 * err * X[i])
        return self
```

The fixed seed makes the random choice of observations the same on every run, so the results repeat.

Extra: Picking observations at random can choose some observations twice in an epoch and others not at all. An alternative is to shuffle the observations once per epoch and walk through them in that order, so each observation is used exactly once. `scikit-learn`'s `SGDRegressor` does this by default: its `shuffle` parameter, `True` unless changed, shuffles the training data after each epoch (`scikit-learn` docs).

4. Faster progress per epoch

Key point: In 40 epochs, SGD reaches test R^2 0.42 on the diabetes data; batch gradient descent only reaches 0.19.

Figure 1 compares the two on the diabetes data, epoch by epoch.

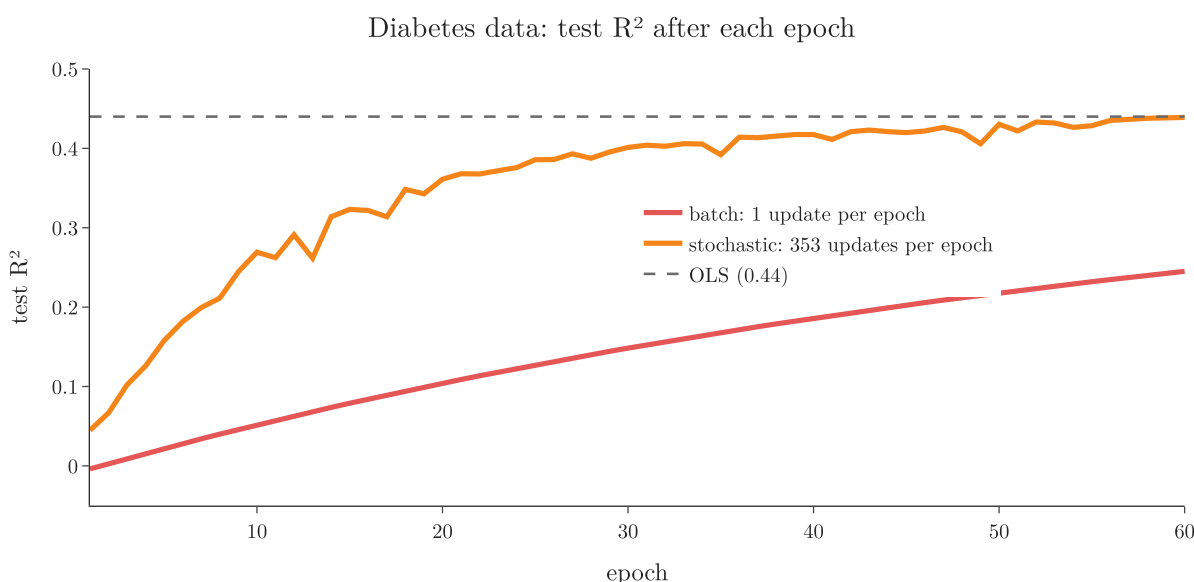


Figure 1: Test R^2 after each epoch: batch vs stochastic

After	Batch (learning rate 0.5)	Stochastic (learning rate 0.01)
10 epochs	0.05	0.27
40 epochs	0.19	0.42
OLS	0.44	0.44

SGD updates 353 times per epoch, batch only once. So per epoch, SGD makes far more progress, even though each of its steps is based on a single observation.

Extra: Per epoch, the Python loop above is slower than batch gradient descent (under a second for 40 epochs against a few milliseconds), because a Python loop over observations is slow and batch uses one fast matrix product. `scikit-learn`'s SGD runs the same loop in compiled code, so it is fast. The real advantage of SGD is elsewhere: one update costs

the same however many observations the data has, so SGD is the main way to train linear models on very large datasets (Goodfellow §5.9).

5. A noisy path

Key point: One observation gives a rough estimate of the whole gradient, so SGD's path zigzags. The noise makes SGD fast and able to escape local minima, but it never settles exactly at the minimum.

5.1 The zigzag

Key point: Each step points roughly, not exactly, downhill.

The gradient from one observation is a noisy estimate of the gradient from all observations. Figure 2 (left) shows the consequence on the 100-point example: batch gradient descent (red) walks straight down the bowl, while SGD (orange) wanders but reaches the bottom area within one epoch.

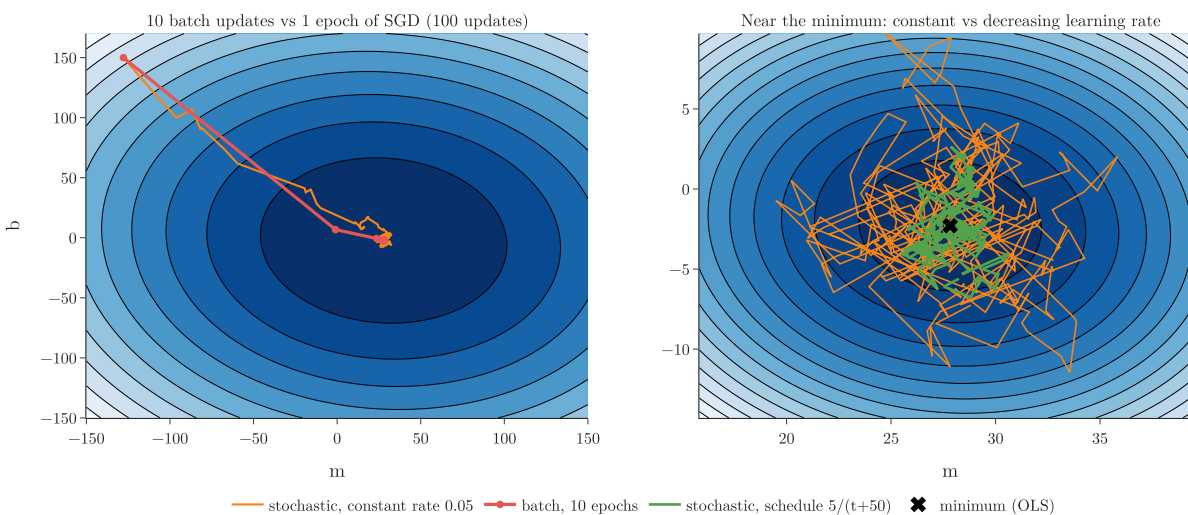


Figure 2: Batch vs stochastic paths, and the effect of a learning schedule

5.2 Advantages of the noise

Key point: The noise speeds SGD up on large data and can shake it out of a local minimum.

- **Large data:** SGD needs only one observation at a time in memory, and it gets close to the answer in few passes.
- **Non-convex losses:** with a loss that has local minima (the gradient descent Note), batch gradient descent settles in whichever dip it reaches first. SGD's random jumps

can carry it out of a sharp, shallow local minimum (Kleinberg et al. 2018), like shaking a tray so a marble hops out of a small dent.

5.3 Disadvantage: jitter at the bottom

Key point: With a constant learning rate, SGD keeps jumping around the minimum instead of stopping at it.

Near the minimum, every new observation still pushes the coefficients in its own direction. So SGD keeps bouncing around the best values (Figure 2, right, orange) and never stops exactly there. Running it twice with different random observations also gives slightly different answers.

6. Learning schedules

Key point: Making the learning rate shrink over time lets SGD move fast at first and settle down near the end.

The fix for the jitter is a **learning schedule**: a learning rate that decreases as training goes on. Big steps early on cover distance quickly; small steps later let SGD settle close to the minimum.

A common schedule is

$$\eta_t = \frac{t_0}{t + t_1}$$

where t counts the updates done so far, and t_0 and t_1 are constants. With $t_0 = 5$ and $t_1 = 50$:

Update t	0	100	1,000	10,000
Learning rate	0.1	0.033	0.0048	0.0005

In Figure 2 (right), the scheduled run (green) stays much closer to the minimum than the constant-rate run (orange): over the last 200 updates its average distance from the best values is 2.1, against 4.7.

Extra: The idea resembles *simulated annealing*, an optimisation method named after annealing metal: cooled slowly, a metal settles into a stable, low-energy state. Simulated annealing likewise lowers a “temperature” step by step (Kirkpatrick et al.).

7. SGD in scikit-learn

Key point: SGDRegressor trains linear regression with stochastic gradient descent; `learning_rate` and `eta0` set the schedule.

Python: scikit-learn's SGDRegressor.

```
from sklearn.linear_model import SGDRegressor

reg = SGDRegressor(max_iter=100, learning_rate="constant",
                   eta0=0.01, random_state=0)
reg.fit(X_train, y_train)
reg.score(X_test, y_test)      #  $R^2$  0.43
```

The main parameters:

Parameter	Meaning
max_iter	the maximum number of epochs
eta0	the starting learning rate
learning_rate	the schedule: "constant", "invscaling" (the default, $\eta_0/t^{0.25}$), "optimal" or "adaptive"
tol	stop early when the loss stops improving by at least this much
random_state	fixes the random order of observations

On the diabetes data, "constant" with $\text{eta0}=0.01$ reaches test R^2 0.43 in 97 epochs. The default "invscaling" schedule shrinks the rate as $\eta_0/t^{0.25}$, so it needs a larger start: with $\text{eta0}=0.2$ it reaches 0.45 in 75 epochs, the best of these runs and above OLS's 0.44. A shrinking schedule is like a car braking as it nears a parking spot: it must start fast enough to arrive at all.

Extra: With the same small start, $\text{eta0}=0.01$, the shrinking schedule stops at test R^2 0.16 after 100 epochs: the steps shrink before the coefficients get near the answer. The same schedule reaches 0.39 after 1,000 epochs and 0.45 after 5,000; switching the shrinking off ($\text{power_t}=0$) gives 0.43 in 100 epochs. Starting rates between: 0.05 gives 0.38, 0.1 gives 0.43 (notebook). Schedules need tuning like any other hyperparameter (Goodfellow §8.3.1).

Extra: scikit-learn may print a `ConvergenceWarning` when `max_iter` runs out before its stopping rule (`tol`) is met. The warning is not an error; its own message suggests increasing `max_iter`.

8. When to use which

Key point: Batch for small data and smooth convex losses; SGD for very large data, data that does not fit in memory, and non-convex losses.

	Batch	Stochastic
Observations per update	all	one
Path	smooth, exact gradient	noisy, zigzags
Progress per epoch	slow	fast
Memory	whole dataset	one observation
Final answer	settles at the minimum	jitters around it (use a schedule)
Local minima	can get stuck	can escape

Mini-batch gradient descent, the next Note, sits between the two and is what most practice uses.

9. Summary

- Batch gradient descent needs about $n \times m$ operations per update: too slow and too memory-hungry for big data.
- SGD updates after each random observation: n updates per epoch, each using one observation's derivatives.
- On the diabetes data, 40 epochs of SGD reach test R^2 0.42, against 0.19 for batch.
- The noise makes SGD fast and able to escape local minima, but it jitters at the bottom.
- A learning schedule (shrinking rate) lets SGD settle, if its starting rate is large enough; `SGDRegressor` offers several.

10. Sources

- **Goodfellow:** I. Goodfellow, Y. Bengio, A. Courville, *Deep Learning*, MIT Press, 2016 (deeplearningbook.org). §5.9 (SGD for large datasets), §8.3.1 (SGD and decaying learning rates).
- **scikit-learn docs:** scikit-learn documentation, `sklearn.linear_model.SGDRegressor` (version 1.9).
- **Kirkpatrick et al.:** S. Kirkpatrick, C. D. Gelatt, M. P. Vecchi, “Optimization by Simulated Annealing”, *Science* 220(4598), 671–680, 1983.
- **Kleinberg et al. 2018:** R. Kleinberg, Y. Li, Y. Yuan, “An Alternative View: When Does SGD Escape Local Minima?”, *Proceedings of the 35th International Conference on Machine Learning (ICML)*, PMLR 80, 2018.

11. Key terms

Term	Meaning
Feature	An input variable: one column of the data table
Observation	One record: one observation of the data table
Target	The output we predict
Stochastic	Involving randomness
Stochastic gradient descent (SGD)	Gradient descent that updates the coefficients after each single random observation
Learning schedule	A rule that changes the learning rate during training, usually shrinking it
SGDRegressor	scikit-learn's linear regression trained with stochastic gradient descent
eta0	The starting learning rate in SGDRegressor
max_iter	The maximum number of epochs in SGDRegressor

Term	Meaning
Simulated annealing	An optimisation method that lowers a “temperature” step by step so the search settles into a low state; the idea behind shrinking learning rates